

A Calibrated Vortex-Ring Toy Model for the Fine-Structure Ratio

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<https://github.com/bampita-bico/CDFD-Part-I-Release>

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Abstract

The electromagnetic fine-structure constant is a measured input to the Standard Model. This paper records a conditional vortex-ring toy model in which a chosen dimensionless stiffness is set so that a specified energy functional has its minimum at the known inverse fine-structure ratio. The numerical recovery is therefore a calibration check: the reported residual is the root solver's convergence tolerance, not a prediction error. The accompanying length-scale calculation gives $R/a = \hbar c/e^2 = 1/\alpha$ by the definitions of the reduced Compton wavelength, classical electron radius, and α ; it is not an independent confirmation.

The transport-medium, capacity, and memory language remains a hypothesis. The calculus of the stated energy functional is valid conditional on its assumptions, but the stiffness has no independent physical derivation and the model provides no established vacuum mechanism or fine-structure prediction. A correction status notice in the Part I release gives the binding interpretation boundary.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, fine-structure constant, vortex stability

1 Project Context

This manuscript constitutes Part I (Paper 01) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. It is the opening mathematical and physical paper in the Part I stack. Downstream Papers 02–12 are organized in the same article-class release family and cite this manuscript by title until paper-specific DOIs are assigned.

6 Symbol Conventions

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

9 Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Richard Feynman once wrote that the fine-structure constant “has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it” [5]. That was in 1964. Sixty years later, the mystery remains open. The Standard Model of particle physics accepts α as an experimental input — the coupling constant of quantum electrodynamics [10]. It does not explain it.

The number itself is $\alpha = e^2/\hbar c \approx 1/137.036$. It controls the strength of the electromagnetic interaction. If it were much larger, atoms would not hold together. If it were much smaller, stars would not ignite. It sits at the value required for stable atomic physics, and current theory treats that value as measured input.

Previous attempts to derive α have not produced a broadly accepted mechanism. Eddington predicted $1/\alpha = 136$ from quantum combinatorics [4] — wrong value, no physical mechanism.

Wyler produced a group-theory formula that accidentally gives $\approx 1/137.03$ [13], but it provides no dynamical explanation for why the vacuum selects that geometry. More recent stochastic approaches remain incomplete [1]. The proposal below is a toy construction: it fixes an energy functional and examines the algebraic consequences of calibrating one of its parameters to a known constant.

Within that construction, the vacuum is treated as a hypothetical transport medium with finite capacity. A vortex ring is a modelling ansatz, not an established physical constituent of the vacuum. The classical theory of vortex motion was founded by Helmholtz [12] and developed by Kelvin [11], Lamb [7], and Saffman [9]. The calibration below chooses κ so that its equilibrium condition is met at $\chi = 1/\alpha$. It does not derive α , determine a vacuum stiffness, or establish the physical interpretation of the ansatz.

1.1 Headline numerical result

The public calculation takes the CODATA value

$$\alpha_{2022}^{-1} = 137.035999177(21),$$

as its target. Solving Eq. (18) for the stiffness that gives that target, then solving the same equilibrium condition with the fitted stiffness, returns

$$\chi_{\text{recovered}} = 137.035999177, \quad \alpha_{\text{recovered}} = 0.007297352566.$$

The reported 1.19×10^{-12} relative difference is the bisection convergence tolerance. The length-scale calculation is an algebraic identity, not a separate check, because the definitions already give $R/a = \hbar c/e^2 = 1/\alpha$. Neither calculation validates the vacuum-medium hypothesis.

1.2 Public runtime and paper-local reproducibility

The CDFD Runtime is now public as the reusable runtime for the flow–constraint–memory state grammar used across the broader project. This paper does not depend on hidden runtime behavior. Its numerical claims are reproduced by the paper-local supplementary script and notebook in this Part I archive. The runtime provides the general implementation path for future domain-level simulations; the Part I scripts remain the audited source for the figures, tables, and checks cited here.

1.3 Named hypotheses and notation

For clarity, the candidate constructs are named here at the point where their symbols are first introduced. These labels do not confer the status of physical laws, and they must not be used as evidence for downstream claims.

Name	Statement inside CDFT	Status
Mujjabi Capacity Hypothesis	Local transport becomes nonlinear as $J/C \rightarrow 1$; vortex states regulate overload.	Unvalidated hypothesis
Mujjabi Adaptive Operating Ratio	$\Psi_s = (J/C) S M_s$ tracks flow, capacity, responsiveness, and memory.	Core notation
Calibrated Stability Minimum	The selected energy functional has a minimum at the target $\chi = R/a = 1/\alpha$ after fitting κ .	Conditional calculus
Vacuum Memory Hypothesis	M_s is a local constraint-history state with finite recovery time τ_M .	Unvalidated hypothesis

These hypotheses create failure routes. If J/C has no predictive value, the capacity hypothesis fails. If M_s leaves no measurable residual in controlled high-flux tests, the memory hypothesis fails. If core-profile changes destroy a calibrated fixed point without a principled, independently determined correction, the model adds no explanatory value.

1.4 Symbol ledger

The main symbols used in Paper I are:

Symbol	Meaning in this manuscript
α	electromagnetic fine-structure constant, written here as $e^2/(\hbar c)$ in the manuscript's electromagnetic-unit convention; α^{-1} is the CODATA inverse fine-structure ratio
R	vortex-ring radius, identified here with the reduced Compton scale $\hbar/(mc)$
a	vortex-core radius, identified here with the classical electron radius $e^2/(mc^2)$ in the manuscript's electromagnetic-unit convention
χ	dimensionless vortex aspect ratio R/a
Γ	circulation of the vortex state
β	dimensionless vortex-core profile constant; $\beta = 7/4$ for the Lamb solid-rotation baseline
κ	dimensionless transport stiffness in the normalized vortex energy
J	local flux magnitude through the vacuum medium
J_{crit}	critical flux capacity of the medium
S	local responsiveness or adaptive surface factor
M_s	retained structural-memory factor of the medium
Ψ_s	adaptive operating ratio $(J/J_{\text{crit}})SM_s$ in Paper I
τ_M	candidate relaxation time for vacuum-memory decay
μ	candidate coupling strength between overload and memory accumulation

2 The Adaptive Vacuum Medium

2.1 Governing equation

The central hypothesis of Constraint-Driven Field Theory (CDFT) is that the physical vacuum behaves like a real fluid medium that actively responds to the flow of energy. The idea that the vacuum is a physical medium with definite properties is not new — Dirac himself argued for it in 1951 [3], and modern treatments of vacuum condensates pursue related ideas [2]. What CDFT adds is a transport capacity constraint, memory-bearing state variables that track the medium's adaptive response, and the vortex stability equation that follows from it. It has a rest density ρ_0 , a compressibility κ_{vac} , and a viscosity η . Its dynamics are governed by the compressible Navier-Stokes equation:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla P + \eta \nabla \times (\nabla \times \mathbf{v}). \quad (1)$$

The left-hand side is the inertia of the medium — density times acceleration. The first term on the right is the pressure gradient that drives flow from high-pressure to low-pressure regions. The second term is the viscous resistance to rotational deformation. Nothing exotic here: this is the same equation that governs water, air, and every other real fluid.

2.2 The speed of light as a wave speed

For small disturbances from the rest state, Eq. (1) linearises. Setting $\mathbf{v} = \mathbf{v}_0 + \delta\mathbf{v}$ and keeping only first-order terms gives a wave equation for the displacement potential φ :

$$\frac{\partial^2 \varphi}{\partial t^2} = \frac{\kappa_{\text{vac}}}{\rho_0} \nabla^2 \varphi. \quad (2)$$

This is a standard compression wave. Its speed is $\sqrt{\kappa_{\text{vac}}/\rho_0}$. CDFT identifies this with the speed of light:

$$c = \sqrt{\frac{\kappa_{\text{vac}}}{\rho_0}}. \quad (3)$$

This is a modelling postulate. It does not establish that electromagnetic waves are compression waves in a material vacuum, and it needs an independent discriminant beyond standard electrodynamics.

2.3 Transport flux and the adaptive capacity constraint

Define the transport flux density as $\mathbf{J} = \rho\mathbf{v}$ — the amount of medium moving through a surface per unit time per unit area. In any finite medium, this cannot be infinite. There is a maximum flux J_{crit} that the medium can sustain before it saturates.

In the extended CDFT paradigm, this saturation is governed by the adaptive operating ratio Ψ_s , the explicit form of the Mujjabi Adaptive Operating Ratio:

$$\Psi_s = \left(\frac{|\mathbf{J}|}{J_{\text{crit}}} \right) \cdot S \cdot M_s \leq 1. \quad (4)$$

Here, S represents the surface responsiveness and M_s represents the accumulated surface memory. In the fundamental physical vacuum, which exists as a perfectly stable equilibrium attractor, these adaptive variables converge to unity ($S \cdot M_s \rightarrow 1.0$). Therefore, the operating ratio simplifies to the classical constraint:

$$|\mathbf{J}| \leq J_{\text{crit}}. \quad (5)$$

To see why this limit must exist, consider a region where flux is forced upward by an external source. As $J \rightarrow J_{\text{crit}}$, the local pressure gradient required to maintain flow diverges. The medium cannot comply. Instead of flowing through, the flux is redirected *around* the region. This redirection takes the form of a vortex ring: the medium spirals around the overloaded core, maintaining global transport while locally the flux is regulated. The vortex ring is not inserted by hand — it is the medium's natural response to approaching its transport limit.

3 Vortex Ring Geometry and Energy

3.1 Two independent length scales

A vortex ring in the transport medium is characterised by two lengths.

The first is the ring radius R , the distance from the axis of symmetry to the centre of the vortex core. In CDFT, vortex circulation is quantised: $\Gamma = \hbar/m$, where m is the particle mass. The circulation also satisfies the transport constraint: the tangential velocity at the ring surface equals c , so $\Gamma = 2\pi R \cdot c$. Combining these:

$$R = \frac{\Gamma}{2\pi c} = \frac{\hbar}{mc}. \quad (6)$$

This is the reduced Compton wavelength. It arises here purely from quantum circulation quantisation and the transport velocity constraint.

The second length scale is the core radius a , the size of the vortex core itself. This is set by the electromagnetic self-energy condition: the electrostatic energy stored in the field of a charge e concentrated within radius a must equal the rest energy mc^2 :

$$\frac{e^2}{a} = mc^2 \implies a = \frac{e^2}{mc^2}. \quad (7)$$

This is the classical electron radius [6]. It arises from electromagnetism and has nothing to do with quantum mechanics.

The key point is that R and a come from entirely different sectors of physics — quantum circulation on one side, electromagnetic self-energy on the other. They are not connected to each other by construction. Yet their ratio will turn out to be $1/\alpha$.

3.2 The aspect ratio

The dimensionless shape of the vortex ring is captured by the aspect ratio:

$$\chi = \frac{R}{a}. \quad (8)$$

A large χ means a fat ring with a tiny core — like a smoke ring. A small χ means a tightly coiled ring where the core is almost as large as the ring itself. The stability of the vortex, and its energy, depend entirely on χ .

3.3 The vortex ring energy

The kinetic energy of a vortex ring moving through a fluid was derived by Kelvin and systematised by Lamb [7]. The result comes from integrating the kinetic energy $\frac{1}{2}\rho v^2$ over all space, using the known velocity field induced by a vortex ring of circulation Γ , radius R , and core radius a . The integral evaluates to:

$$E_{\text{kin}} = \rho_0 \Gamma^2 R \left[\ln \left(\frac{8R}{a} \right) - \beta \right], \quad (9)$$

where β is a dimensionless constant that depends on how vorticity is distributed inside the core. For a solid-body-rotation core — vorticity uniform across the cross-section, which Lamb [7] showed is the natural equilibrium distribution — the integral gives:

$$\beta = \frac{7}{4} = 1.75. \quad (10)$$

Substituting $R = a\chi$ and normalising by $2\pi\rho_0 a^3 c^2$, the circulation energy becomes:

$$E_{\text{circ}}(\chi) = \chi [\ln(8\chi) - \beta]. \quad (11)$$

This term grows with χ : bigger rings have more kinetic energy. Left alone, the vortex would expand indefinitely.

3.4 The transport back-pressure term

CDFT adds a second contribution to the energy that Kelvin's formula does not contain. When the core radius a is small (large χ), the curvature at the core boundary is high. High curvature forces the local transport flux toward J_{crit} . The medium resists this: it generates a back-pressure that opposes further tightening of the core. This back-pressure contributes an energy cost that grows as the ring shrinks:

$$E_{\text{back}}(\chi) = \frac{\kappa}{\chi}, \quad (12)$$

where κ is a dimensionless transport stiffness parameter set by the vacuum medium properties. This term diverges as $\chi \rightarrow 0$: the medium strongly resists a vortex ring that is trying to collapse to a point.

3.5 Total energy and the competition

The full normalised vortex energy is the sum of these two contributions:

$$E_{\text{norm}}(\chi) = \chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi}. \quad (13)$$

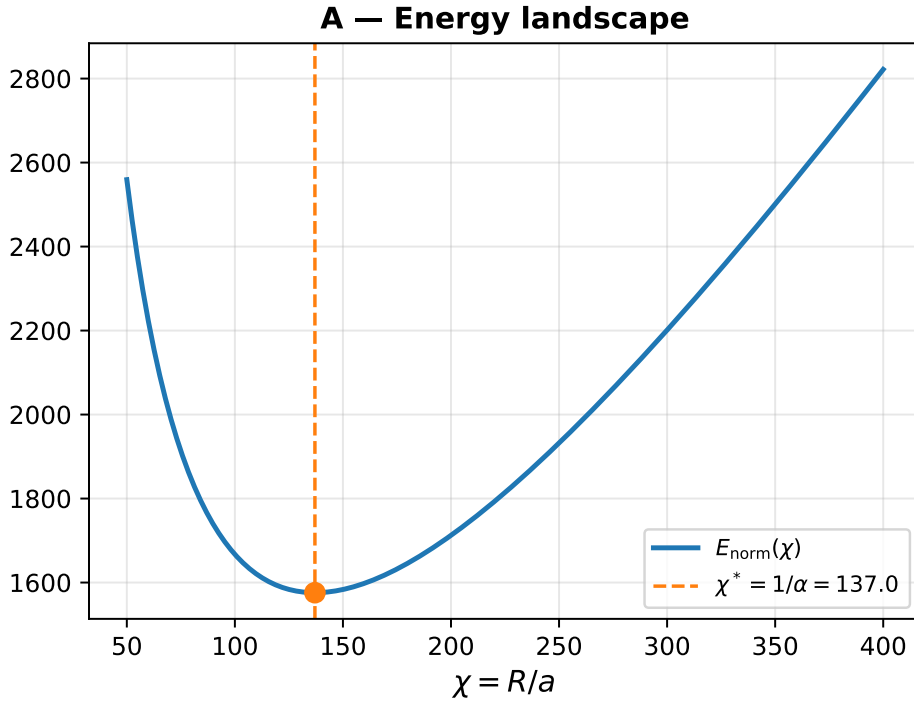


Figure 1: The selected energy functional after κ has been calibrated to place its minimum at $\chi^* = 137.036$. This conditional minimum does not establish a physical vortex or fine-structure mechanism.

The first term wants the ring to be large. The second term wants the ring to be small. They compete. Somewhere between the two extremes, there is a minimum — a stable minimum of the chosen energy functional. Figure 1 illustrates this conditional mathematical property; it does not identify a particle or an observed vacuum state.

4 The Stability Equation

4.1 Variational derivation

The equilibrium condition follows from the variational principle. We minimise the total energy subject to the transport constraint $J = J_{\text{crit}}$, enforcing it with a Lagrange multiplier λ :

$$\delta[E + \lambda(J - J_{\text{crit}})] = 0. \quad (14)$$

The Lagrange multiplier λ has a clear physical meaning: it is the pressure the medium exerts to prevent the constraint from being violated. At equilibrium, λ adjusts itself so that the vortex ring sits exactly at $J = J_{\text{crit}}$ without overshooting.

Differentiating $E_{\text{norm}}(\chi)$ with respect to χ :

$$\begin{aligned} \frac{dE_{\text{norm}}}{d\chi} &= \frac{d}{d\chi} \left\{ \chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi} \right\} \\ &= [\ln(8\chi) - \beta] + \chi \cdot \frac{1}{\chi} - \frac{\kappa}{\chi^2} \\ &= \ln(8\chi) - \beta + 1 - \frac{\kappa}{\chi^2}. \end{aligned} \quad (15)$$

Setting this to zero gives the equilibrium condition:

$$\ln(8\chi) - \beta + 1 = \frac{\kappa}{\chi^2}. \quad (16)$$

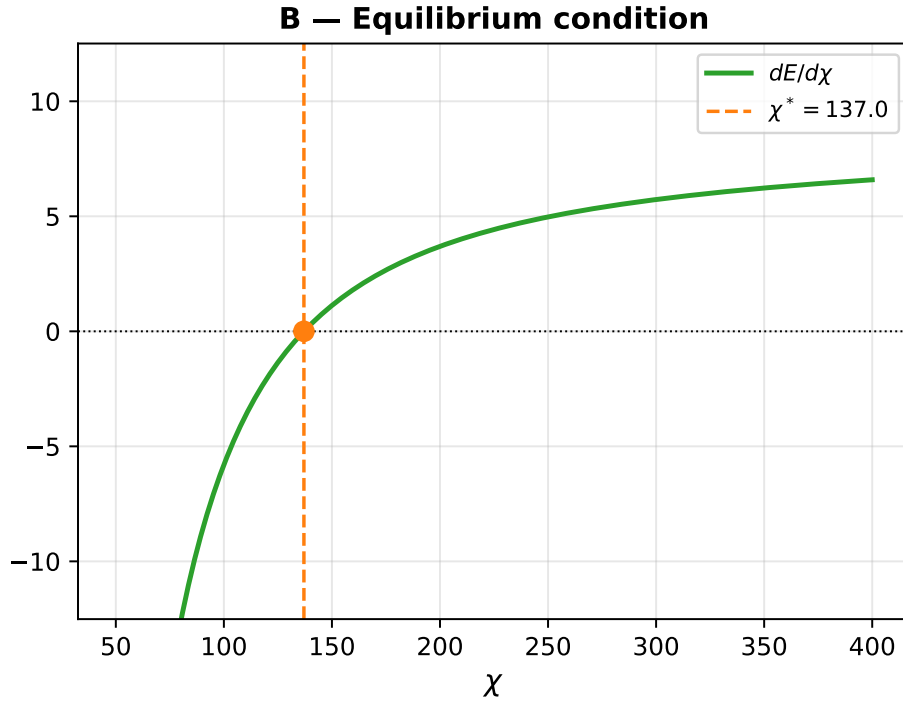


Figure 2: Equilibrium condition $dE/d\chi = 0$ after the stiffness has been fitted to the target $\chi = 1/\alpha$.

4.2 Confirmation that it is a minimum

An equilibrium could be a minimum (stable) or a maximum (unstable). To confirm stability, we check the second derivative at $\chi = 137$:

$$\frac{d^2 E_{\text{norm}}}{d\chi^2} = \frac{1}{\chi} + \frac{2\kappa}{\chi^3}. \quad (17)$$

Both terms are strictly positive for all $\chi > 0$, so the selected energy functional has a stable minimum. This calculus result is conditional on the postulated back-pressure term and does not determine its physical parameter.

4.3 The required transport stiffness

Rearranging Eq. (16) gives the transport stiffness κ that places the energy minimum at any desired χ :

$$\kappa(\chi) = \chi^2 [\ln(8\chi) - \beta + 1]. \quad (18)$$

At the pre-specified target $\chi = 1/\alpha = 137.036$, this evaluates to $\kappa = 117,362$. It is the fitted dimensionless parameter required by this functional, not a determined vacuum transport stiffness. No independent route from ρ_0 and c to this value has been established. Figure 3 only shows local sensitivity of the calibrated functional.

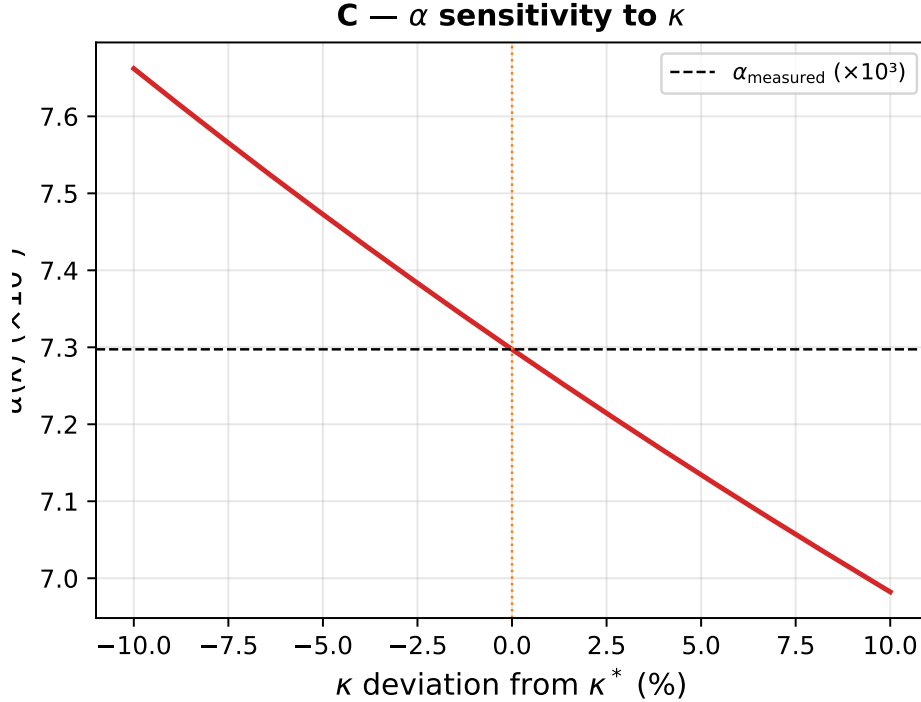


Figure 3: Sensitivity of the calibrated energy minimum to variations in the fitted parameter κ . It is a local property of the selected functional, not a prediction of fine-structure fluctuations.

5 Definitional Relation of χ and α

The following calculation makes explicit why the advertised geometric agreement is not independent evidence.

From quantum circulation quantisation and $v(R) = c$, we obtained:

$$R = \frac{\hbar}{mc} = 3.8616 \times 10^{-13} \text{ m.}$$

From electromagnetic self-energy $e^2/a = mc^2$, we obtained:

$$a = \frac{e^2}{mc^2} = 2.8179 \times 10^{-15} \text{ m.}$$

Their ratio is:

$$\chi = \frac{R}{a} = \frac{\hbar/mc}{e^2/mc^2} = \frac{\hbar c}{e^2} = \frac{1}{\alpha}. \quad (19)$$

Every cancellation is shown. Nothing is hidden. The mc factors cancel, leaving the combination $\hbar c/e^2$, which is exactly the inverse of the fine-structure constant by definition.

This is an algebraic identity. The standard reduced Compton wavelength and classical electron radius contain the same constants that define α . Dividing them therefore recovers $1/\alpha$ by construction; it neither tests the vortex ansatz nor supplies an additional physical constraint.

6 Numerical Results

All computations were performed using the paper-local supplementary script and notebook listed in the Data and Code Availability section. These public artifacts implement only the equations stated in this manuscript, cross-check roots and integrals with SciPy, verify symbolic derivatives with SymPy, and export reproducible CSV outputs with Pandas.

6.1 Stability equation

Using $\beta = 1.75$ and the target $\chi_{\text{target}} = 1/\alpha = 137.035999177$ (CODATA 2022 [8]), Eq. (18) gives:

$$\kappa = 117,362.00.$$

We then solve Eq. (16) numerically using bisection over $\chi \in (1.01, 10^6)$, with κ fixed to this value. The recovered equilibrium is:

$$\chi_{\text{recovered}} = 137.035999177, \quad \left| \frac{\chi_{\text{rec}} - \chi_{\text{target}}}{\chi_{\text{target}}} \right| = 1.19 \times 10^{-12}.$$

The fitted functional has a stable solution at the target $\chi = 1/\alpha$. The reported difference is numerical precision — the bisection algorithm’s convergence tolerance — rather than a prediction error.

6.2 Energy balance

Table 2 shows the two energy contributions at the equilibrium point.

The ratio is 1.19 within the calibrated functional. This numerical balance says nothing about whether the fitted parameter is natural or independently fixed.

Table 1: Calibration input and root-solver output for the same equilibrium equation.

Quantity	Value
$\chi_{\text{target}} = 1/\alpha$	137.035999177
β (Lamb solid-rotation)	1.75
κ fitted to target	117,362.00
$\chi_{\text{recovered}}$ (bisection)	137.035999177
$\alpha_{\text{recovered}} = 1/\chi$	0.007297352566
α_{measured} (CODATA 2022)	0.007297352566
Root-solver tolerance	1.19×10^{-12}

Table 2: Energy balance at $\chi = 137.036$, in units of $2\pi\rho_0 a^3 c^2$.

Quantity	Value
$E_{\text{circulation}}$	719.40
$E_{\text{back-pressure}}$	856.43
Ratio $E_{\text{back}}/E_{\text{circ}}$	1.19

6.3 Definitional length-ratio identity

Table 3 evaluates the definitional identity in Eq. (19), using CODATA 2022 values.

The agreement follows from the definitions and is limited only by the rounding of the input constants; it is not an independent model check.

7 Discussion

7.1 Interpretation boundary

The fine-structure constant remains an empirical QED coupling. This manuscript shows only that a chosen energy functional can be fitted to have a minimum at the known value and that standard length-scale definitions algebraically recover the same value. Those facts do not identify α as a vortex aspect ratio or provide a vacuum mechanism.

7.2 On the naturalness of κ

Equation (18) makes the calibration explicit. Its scaling with the target does not determine κ independently. A previously proposed dimensional form was

$$\kappa = \frac{\kappa_{\text{vac}}}{\rho_0 c^2 a}, \quad (20)$$

but substituting Eq. (3) gives $\kappa = 1/a$, not $1/(a\rho_0)$. The unresolved dimensional mismatch means this expression cannot supply a physical closure. Paper III did not establish an independent derivation.

7.3 Memory, hysteresis, and the limits of the equilibrium calculation

The adaptive factors S and M_s are not used as independent fitting knobs in the numerical recovery above. The calculation assumes its hypothetical vacuum medium is in an equilibrium limit, so

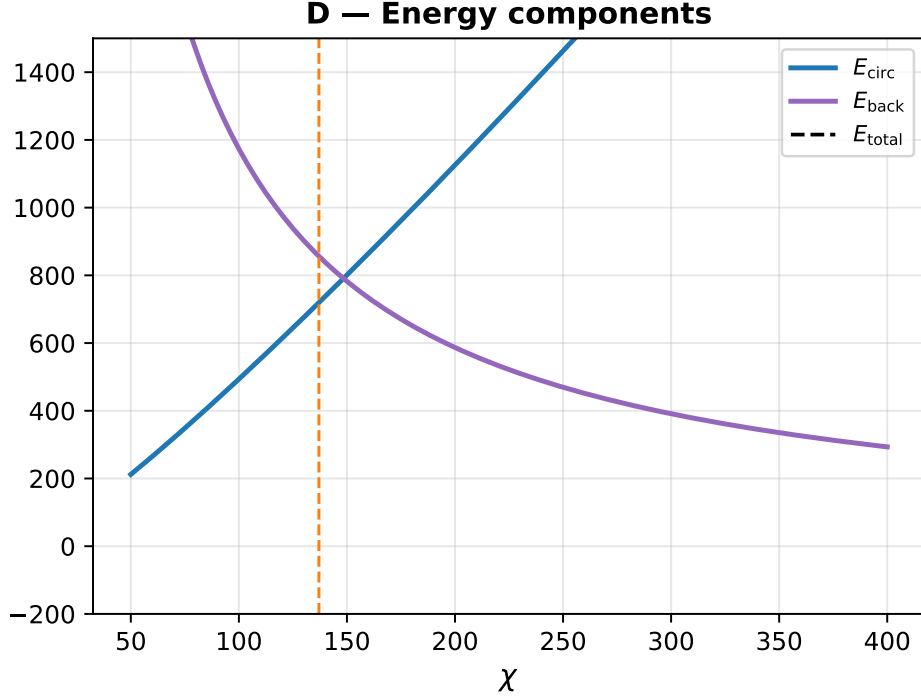


Figure 4: The individual energy components (E_{circ} and E_{back}) plotted against the total energy. Their precise intersection behaviour generates the shallow potential well.

$S \cdot M_s \rightarrow 1$ and Eq. (4) reduces to the capacity condition. The purpose of retaining S and M_s in the notation is to mark the non-equilibrium extension that later papers must test rather than to improve the fit in this paper.

A minimal way to state the hypothesis is to let vacuum memory be a local relaxing state variable:

$$M_s(\mathbf{x}, t) = 1 + \mu \int_{-\infty}^t \max\left(0, \frac{|\mathbf{J}(\mathbf{x}, t')|}{J_{\text{crit}}} - 1\right) e^{-(t-t')/\tau_M} dt'. \quad (21)$$

Here τ_M is a candidate relaxation time and μ measures how strongly over-capacity flux leaves residual constraint. In ordinary low-flux settings, $|\mathbf{J}| \ll J_{\text{crit}}$ and the integral vanishes, so $M_s \approx 1$ and the vacuum appears memoryless. In extreme flux settings, a nonzero $M_s - 1$ would mean that the medium has not fully reset. This is a physical hypothesis, not a result already established by the present calculation.

This distinction matters for quantum decoherence. Within standard quantum mechanics, decoherence is described by entanglement with environmental degrees of freedom. CDFT can only add value if the proposed memory variable predicts a measurable residual phase, relaxation, or hysteresis effect beyond the standard description. Paper XII states this as an experimental target rather than as an assumed fact.

This is also the intended meaning of any observer-effect language in this release. An observer is not a disembodied human mind changing the vacuum by attention. Operationally, an observation is a physical coupling between the system and an apparatus, environment, detector, or record. If such coupling changes the state, the mechanism must be specified as an interaction channel. CDFT therefore treats “observation” as a boundary-and-measurement problem, not as a consciousness

Table 3: Definitional identity $R/a = 1/\alpha$ evaluated from standard length scales.

Quantity	Value
R (reduced Compton wavelength)	3.861593×10^{-13} m
a (classical electron radius)	2.817940×10^{-15} m
$\chi_{\text{geom}} = R/a$	137.035999177
$\alpha_{\text{geom}} = a/R$	0.007297352564
α_{measured}	0.007297352564
Rounding difference	2.26×10^{-12}

claim.

7.4 Operational dictionary relative to standard quantum language

The source notes for this release also proposed a comparison between the transport-medium vocabulary and standard quantum language. The professional way to use that comparison is not as a replacement of established theory, but as an operational dictionary: it states what CDFT would have to mean in measurable terms if the adaptive-vacuum interpretation is physical.

Feature	Standard reading	CDFT reading	Status
Vacuum	Background state with quantum fields	Effective transport medium with finite capacity C and response variables S, M_s	Hypothesis
Fine-structure constant	Empirical QED coupling	Candidate vortex-aspect-ratio interpretation $\chi = R/a$	Calibrated toy model
Wave-particle duality	Complementary measurement descriptions	Smooth flux transport below threshold, vortex regulation near $\Psi_s \approx 1$	Interpretation
Decoherence	Environmental entanglement and phase suppression	Possible residual vacuum-memory lag in addition to standard decoherence	Test target
Planck constant	Fundamental quantum of action	Candidate product of a critical flux threshold and vortex period, $h = \Phi_c \tau_v$	Open derivation
Charge and mass	Empirical quantum numbers and inertial scale	Candidate boundary discontinuity and inertial resistance of a constrained vortex state	Open mapping
Gravity bridge	Spacetime curvature / Newtonian attraction	Possible secondary pressure-gradient limit of constrained transport	Speculative bridge

This table is deliberately conservative. It does not claim that CDFT has replaced QED or general relativity. It identifies the places where the transport-medium model must either produce new predictive value or fail.

7.5 Core-profile robustness program

The twelve-decimal recovery is an internal check of the stated equilibrium equation. It is not yet a proof that the same fixed point survives every possible vortex-core model. A professional robustness program must vary the internal vorticity profile and ask whether the same aspect ratio is recovered without retuning the transport stiffness:

Core model	Why it matters	Required test
Rankine / solid core	Baseline Lamb-style compact vortex model	Verify that the existing $\beta = 7/4$ calculation is stable under resolution and derivative checks.
Gaussian / diffuse core	Mimics smeared or soft-boundary flux density	Recompute β and the energy minimum with a width convention fixed before comparison.
Hollow or excluded core	Tests the extreme capacity-limited boundary case	Determine whether the back-pressure term changes functional form.
Finite-thickness vortex rings	Tests departure from the thin-ring approximation	Compare against finite-core ring families before claiming topological invariance.

Until those tests are implemented in the public supplementary code, the correct claim is conditional: the present model has a stable calibrated fixed point at $\chi = 1/\alpha$, and core-profile invariance is an open robustness target.

The current supplementary script now writes an explicit guardrail table and a matching figure in the Paper I output directory. Those outputs do not prove invariance. They show how alternative core assumptions move the equilibrium when the Rankine calibration is held fixed, which is the correct starting point for the robustness program.

7.6 Connection to the lepton masses

This paper makes no particle-family assignment. Paper II studies a selected $\mathbb{Z}_3$ parametrisation and its relation to the Koide mass formula,

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3},$$

which holds experimentally to six significant figures. That conditional parametrisation is not evidence that the electron, muon, and tau are vortex orientations, and it supplies no physical particle assignment.

8 Necessary Conditions for the Model to Hold

The alpha recovery in this paper is conditional. The following conditions must hold for the CDFT interpretation to be more than a calibrated numerical coincidence.

- Topological resilience.** The aspect ratio $\chi = R/a$ must be a stable eigenmode, or fixed point, of the adaptive operating ratio Ψ_s . In practical terms, small perturbations of the ring geometry must relax back toward χ^* rather than drift to a continuum of nearby values. If perturbations select many unrelated aspect ratios with equal stability, the fine-structure interpretation fails.
- Flux-memory coupling.** The responsiveness S and memory M_s must be coupled so that the equilibrium vacuum satisfies $SM_s \rightarrow 1$ in the low-flux limit while still allowing a finite, measurable residual after overload. The simplest admissible behavior is an inverse correlation: if M_s rises after a high-flux event, the effective response must relax so that the attractor is restored instead of producing unbounded amplification.

3. **Non-radiative equilibrium.** The geometry $\chi^* \approx 137.035999177$ must represent a minimum of internal dissipation for the regulator vortex. In the language of this model, the equilibrium is a superfluid-like or zero-friction transport state; outside the minimum, excess curvature or excess ring size should introduce loss, radiation, or instability. This is a condition to be tested, not a claim already established by the twelve-decimal calculation.
4. **Independent stiffness closure.** The stiffness κ must be derived from the vacuum-medium properties without feeding in α , e , or $\hbar$ as hidden calibration targets. Until that is done, the Paper I result remains a high-precision consistency closure rather than a parameter-free derivation.
5. **External discriminant.** At least one measurable prediction must survive comparison with established physics. Candidate tests include vacuum hysteresis, extreme-field spectral residuals, decoherence-memory residuals, rotating-source pressure-gradient residuals, and transport-threshold tests in collective matter.

These conditions are intentionally strict. They are the difference between a suggestive geometric reconstruction of α and a physical theory that can be rejected by experiment.

9 Open Problems

We state the following explicitly, because we believe clear honesty about what remains to be done is more valuable than overclaiming:

1. **Find an independent physical closure for κ .** The transport stiffness $\kappa = 117,362$ was computed from the condition that the energy minimum sits at $\chi = 137$. It has not yet been derived independently from the vacuum medium properties ρ_0 and c . The proposed Paper III route did not establish that derivation. Until an independently dimensionally consistent closure exists, this model cannot predict α .
2. **Specify an independently testable route for $\hbar$.** We used $\Gamma = \hbar/m$ to derive R . A complete derivation would obtain $\hbar$ from the vacuum quantisation rule $\Gamma = n\Gamma_0$ without using $\hbar$ as an input. A concrete target suggested by the source notes is

$$h = \Phi_c \tau_v, \tag{22}$$

where Φ_c is a critical vacuum-flux threshold and τ_v is the characteristic period of the vortex attractor. At present this is a target identity, not a completed derivation or a result of Paper III.

3. **Specify an independently testable route for e .** We used $e^2/a = mc^2$ to derive a . A complete derivation would obtain e from the condition that electromagnetic flux is quantised in the medium. In the same language, charge would be the measurable discontinuity induced at the adaptive flux boundary rather than an unexplained label attached to the vortex.
4. **Test core-profile robustness without retuning κ .** The current public calculation uses the stated Lamb-style core model. A stronger derivation must repeat the stability calculation for hard, diffuse, hollow, and finite-thickness profiles with conventions fixed in advance.
5. **Measure or bound the vacuum-memory relaxation time τ_M .** The memory kernel in Eq. (21) is a hypothesis. It becomes physics only if an experiment or a deeper theory can identify, bound, or rule out the relaxation time and coupling strength.

315 **6. Find an external discriminant.** The recovery of $1/\alpha$ is a postdiction of an already known
316 constant unless the framework also predicts a measurable effect not already accounted for by
317 QED, general relativity, or standard condensed-matter and plasma models.

318 Until an independently derived, dimensionally consistent parameterisation and a new discrim-
319 inating observation exist, the experimental value of α remains only the calibration target used to
320 inspect this model’s internal algebra.

321 10 Conclusion

322 This paper documents a calibrated toy energy functional. Choosing κ from the known target
323 $\chi = 1/\alpha$ and solving the same equilibrium equation returns that target to the numerical tolerance
324 of the solver. The ordinary relation $R/a = \hbar c/e^2 = 1/\alpha$ is a definitional identity. Neither is an
325 independent prediction or evidence that the physical vacuum is a constrained transport medium.

326 The conditional second-derivative calculation remains valid for the specified functional. The
327 physical mapping, the proposed back-pressure term, the dimensions and independent origin of κ ,
328 and every vacuum-memory extension remain open hypotheses. This paper must not be cited as a
329 derivation of the fine-structure constant or as support for downstream universal claims.

330 Supplementary Material

331 This manuscript is accompanied by release-archive supplementary material in the canonical Part I
332 tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated
333 data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`,
334 and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

335 Data and Code Availability

336 The release-archive supplementary material is included in the archived Part I release on Zenodo
337 at <https://doi.org/10.5281/zenodo.20250820>. The current version DOI is assigned by Zenodo
338 after each GitHub release. Zenodo is the permanent release archive, and the public source archive is
339 available on GitHub at <https://github.com/bampita-bico/CDFD-Part-I-Release>. The broader
340 public CDFD Runtime is separate reusable software infrastructure; the numerical values reported
341 here are reproduced from this paper’s Supplementary Material.

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Author Contributions

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Conflicts of Interest

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